Faculty of Information Technology
20241

ASPU
Al-Sham Private University

Multimedia-Lecture-Six

Video

Prof. Houssam AL-Fawal
Eng. Noor AL-Hakim

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Video

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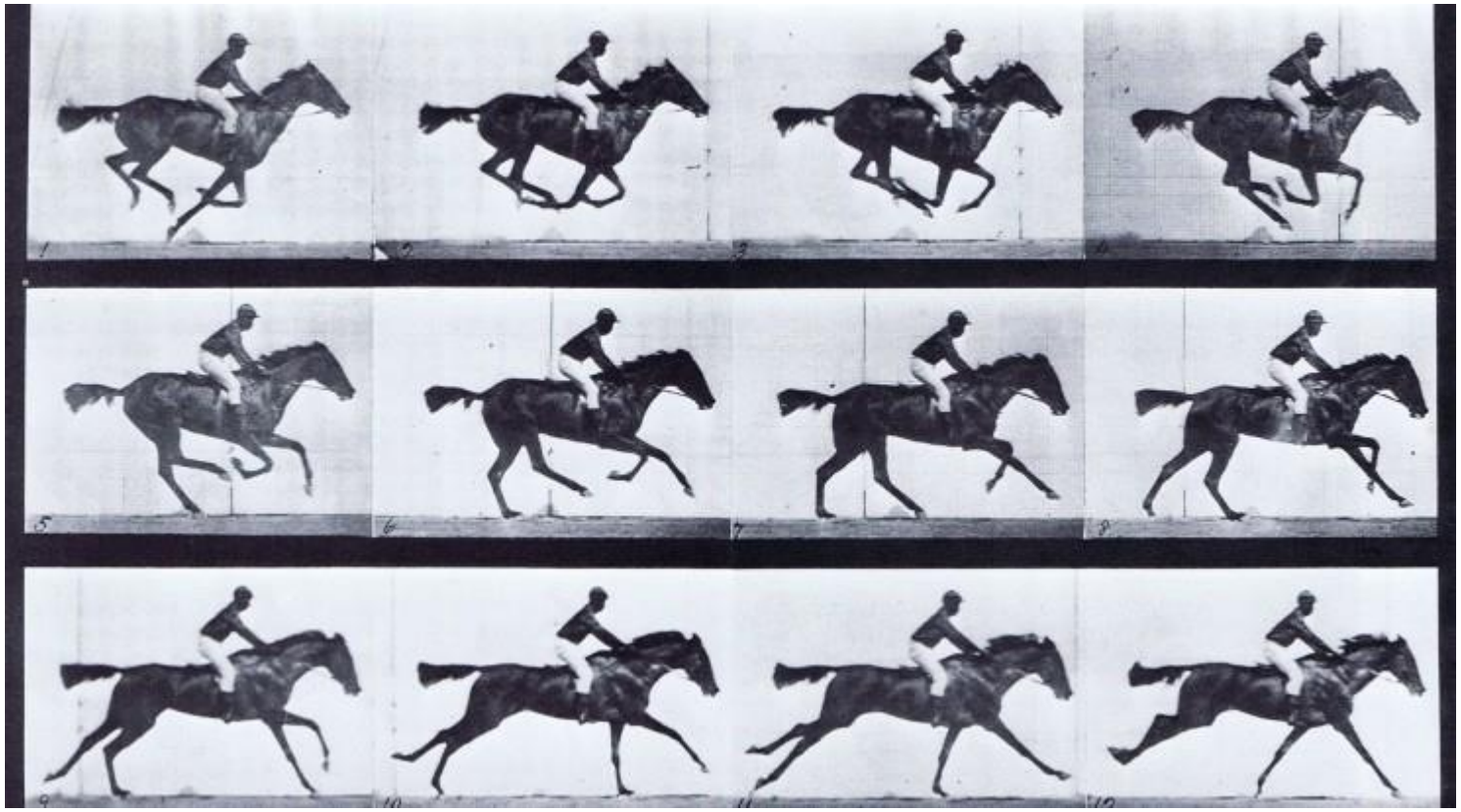
Video Concept

video is a series of digital signals, which simulate movement.



Dealing With Video in PYTHON

- Read a Video File
- Find More information about Video
- Read Frame
- Read Video Frames Starting At Specific Time
- Create VideoWriter Object and Write Video

Read a Video File:

```
>> v = cv2.VideoCapture('myVideo.mp4')
```

The VideoCapture object has properties that contain information about the video file. You can view the value of a property after creating the object.

Find More information about Video:

```
>> fps = v.get(cv2.CAP_PROP_FPS);
```

Fps: Frame per second

Read Frame:

```
# Video capture object
v = cv2.VideoCapture('myVideo.mp4')
# Looping for frames
while True:
  #Reading the next frame
   ret, frame = v.read()
   if not ret:
      break  # Break the loop if the video is over
   # Display the frame
   cv2.imshow('Frame', frame)
  # Break the loop if 'q' is pressed
   if cv2.waitKey(25) & 0xFF == ord('q'):
      break
v.release()
cv2.destroyAllWindows()
```

Read Video Frames Starting At Specific Time:

```
video_path = 'myVideo.mp4'
start_time = 15

cap = cv2.VideoCapture(video_path)
# Set the starting frame index
fps = cap .get(cv2.CAP_PROP_FPS)
start_frame = int(fps * start_time)
cap.set(cv2.CAP_PROP_POS_FRAMES, start_frame)

while cap.isOpened():
    ret, frame = cap.read()
    if not ret:
        break
    cv2.imshow('Frame', frame)
    if cv2.waitKey(25) & 0xFF == ord('q'):
        break
cap.release()
cv2.destroyAllWindows()
```

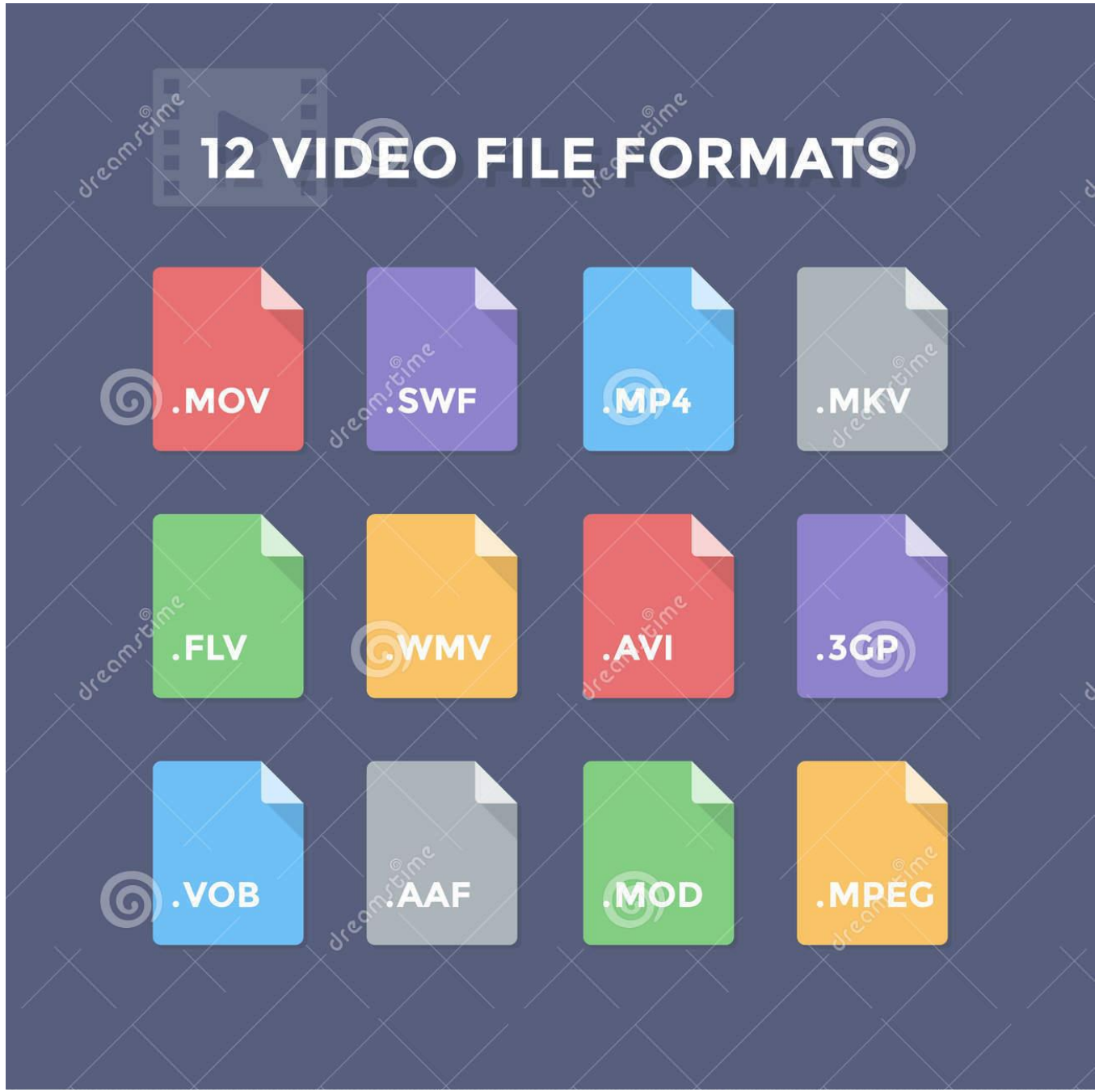
Create VideoWriter Object and Write Video

```
# Specify the encoding algorithm of video
>> fourcc = cv2.VideoWriter_fourcc('M','J','P','G')

# Create a VideoWriter object
>> out= cv2.VideoWriter('output_video.avi', fourcc, fps, (width, height))

# Write the frame to a VideoWriter object
>>out.write(frame)
```

Video Format



Exercise:

I. Write a Python-code to:

a. Read a video file.

b. Reverse frames of that video.

c. Write a reserved video on your disk.

Remain RGB as it's, only reserve the frames

That’s All